

Lesson 3-1

The Overall Structure of Web Applications

Learning Objectives

- Explain what a web application is and describe the roles of its three tiers: frontend, backend, and database.
- Trace how the three tiers cooperate to turn a user's action into a result, using familiar services as examples.

Every time you search on YouTube, buy on Amazon, or open Gmail, you are using a web application — a program you use through your browser, with nothing to install. Behind the simple screen, three tiers work together: the page you interact with, the hidden server that processes the request, and the database that holds all the data. Understanding these three tiers explains how almost every online service is built.

? Guiding question: How is a web application structured, and how do its three tiers work together to respond to what a user does?

Point!

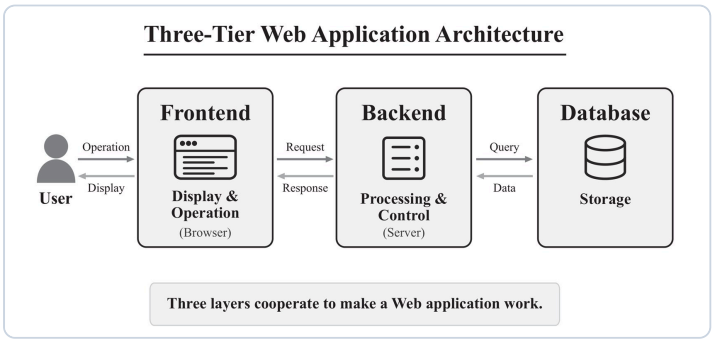
1. What is a Web Application?

(1) **(Web application)**…an application used through a web browser. It can be used as long as you have a browser, without needing to install dedicated software.
e.g., YouTube, Amazon, Gmail, Google Maps

2. The Three Tiers of a Web Application

(1) A web application is broadly composed of the following three tiers.

Tier	Role	Examples
Frontend	The screen component that users see directly. It runs on the browser and accepts user input.	Search screen, product listing pages, input forms, etc.
Backend	The server-side processing that users do not see. It processes data and makes decisions.	Searching for products that match search criteria, login authentication, etc.
Database	A system that organizes, stores, and manages data. It retrieves data when needed and records new data.	Product information, user information, order history, etc.



Three-Tier Web Application Architecture

3. The Flow of How the Three Tiers Cooperate

(1) When a user operates a web application, the three tiers cooperate to perform the processing to make it work.

Key Fact:
A web application is built from three tiers — frontend, backend, and database — that cooperate to turn a user's action into a result.

Key Concepts:

- web application
- frontend
- backend
- database
- three-tier architecture

KEY TERMS
web application — an application used through a web browser, with no dedicated software to install.



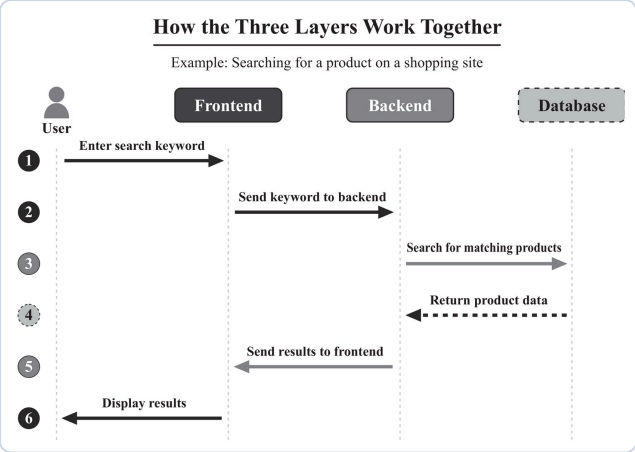
The data tier stores and serves the records.



Three tiers: presentation, logic, and data.

e.g., Searching for a product on an online shopping site

① The frontend receives the search keywords entered by the user.
② The frontend sends the search keywords to the backend.
③ The backend searches the database for products that match the criteria.
④ The database returns the data of matching products to the backend.
⑤ The backend organizes the search results and returns them to the frontend.
⑥ The frontend displays the search results on the user's screen.



How the Three Layers Work Together

4. Familiar Web Application Configurations

(1) All familiar web applications also operate on this three-tier architecture.

Service	Frontend (display and interaction)	Backend (processing and control)	Database (storage and management)
Video sharing site	Display of video playback and search screens; reception of user operations	Selection of recommended videos; aggregation of view counts	Storage of video data, user information, and viewing history
SNS	Display of timelines and post screens; sending of input content	Determining the audience for posts; control of push notifications	Storage of post data, follow relationships, and messages
E-commerce site	Display of product listings and cart screens; reception of order operations	Stock checking; execution of payment systems	Storage of product information, order history, and customer information

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A The frontend is the screen component that users see directly.

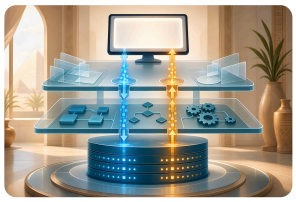
B The backend is the part that runs on the browser.

C A database is a system that organizes, stores, and manages data.

D A web application operates with the frontend alone.

PAUSE & THINK

When you type in a search box and see results, which tier collected your input, and which tier found the answer?



A request flows down the tiers and a reply returns.

KEY TERMS

frontend — the screen users see and interact with.
backend — the hidden server-side processing.
database — where data is stored and managed.

PAUSE & THINK

In the table, every service uses the same three tiers — what changes from one service to the next?

(2) Match each description (a–c) with the most appropriate term from the choices below (A–C).

a The part responsible for server-side processing that users do not see

b The screen component that users interact with directly

c A system that organizes, stores, and manages data

[Choices] **A** Frontend **B** Backend **C** Database

✓ Solution

(1)

A The frontend is the screen component that users see directly. Therefore, ○.

B The backend is the part that runs on the server, not on the browser. Therefore, ×.

C Correct definition of a database. Therefore, ○.

D A web application operates with three tiers: frontend, backend, and database. Therefore, ×.

(2)

a: Server-side processing is the backend. **B**

b: The screen that users interact with is the frontend. **A**

c: Storing and managing data is the database. **C**

Try

1 Answer the following questions.

1. What is the term for an application used through a web browser?
2. What is the term for the screen component that users see directly?
3. What is the term for the part responsible for server-side processing that users do not see?
4. What is the term for a system that organizes, stores, and manages data?

2 Answer the following questions.

(1) When searching for a product on an online shopping site, which of the following is responsible for searching through data for products that match the search keywords: **A** Frontend, **B** Backend, **C** Database?

EXAM-STYLE QUESTION

Explain how the three tiers of a web application cooperate when a user posts a photo on a social media app. **[6] Refer to:** the role of the frontend, backend, and database.

(2) For each of a–f, indicate which tier of an e-commerce site it belongs to: **A** Frontend, **B** Backend, **C** Database.

a Product listing page

b Stock checking processing

c Storage of customer information

d Cart screen

e Payment processing

f Storage of order history

⚙ Think as an Engineer — Investigate & Decide

A clinic wants a web app where patients book appointments, check which time slots are free, and see their past visits. You plan its three tiers.

☐ **Put everything in one program**

☐ **Separate the frontend, backend, and database**

1 · Assign the work. For three features — the booking form, checking which time slots are free, and storing past visits — say which tier (frontend, backend, or database) should handle each, and why.

2 · Diagnose a fault. The booking page opens normally, but no free slots appear on screen. Which tier is clearly working, and which one has probably failed? Explain your reasoning.

3 · Decide. Recommend whether the clinic should separate its app into three tiers, and give two reasons. **In pairs,** compare your assignments and agree together on where checking free time slots belongs.

Give evidence for your answer.

Stuck? The frontend shows and collects; the backend decides and processes; the database stores and retrieves.

🌍 In a New Context

A weather app lets users type a city and see the forecast. Using the three-tier model, name which tier does each job — taking the city name, looking up the forecast data, and storing past forecasts — and identify one job the frontend cannot do on its own.

KEY TERMS

three-tier architecture — frontend + backend + database cooperating.

❓ Lesson Question — Answered

The lesson asked how a web application is structured and how its tiers cooperate. A web application runs in a browser and is built from three tiers. The frontend is the screen the user sees and interacts with; the backend is the hidden server-side processing that makes decisions; and the database organizes, stores, and manages the data. When a user acts — for example, searching for a product — the frontend collects the input and passes it to the backend, the backend queries the database, the database returns the data, and the backend sends the organized result back to the frontend to display. Overall, a web application works not through one part but through three tiers cooperating, each with a clear role.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

A web application is broadly composed of three tiers. The screen component that users see directly is called the (a). The part responsible for server-side processing that users do not see is called the (b). A system that organizes, stores, and manages data is called the (c). In response to user actions, these three tiers cooperate to perform processing.

1. Fill in the blanks (a)–(c).

3 Answer the following questions.

(1) For each of a–f, indicate which tier of an SNS it belongs to: **A** Frontend, **B** Backend, **C** Database.

a Timeline screen
b Processing that determines the audience for a post
c Storage of post data
d Post screen
e Notification processing
f Storage of follow relationships

⭐ REFLECT & CHALLENGE

Reflect: Which tier do you think fails most noticeably to a user when something goes wrong, and why? **Challenge:** Name one online service you use and describe what its frontend, backend, and database each do.

⭐ Key takeaway

Remember:
Frontend shows and collects, backend decides and processes, database stores and retrieves — three tiers cooperating make a web application work.

(2) From the following options (A–D), choose the one that most appropriately describes the three-tier architecture of a web application.

A The frontend has the role of storing and managing data.

B The backend is the part that users interact with directly on the browser.

C In response to user actions, the frontend, backend, and database cooperate to perform processing.

D The database has the role of displaying screens to users.



Learning Objectives

- Explain the client and server model and the difference between HTTP and HTTPS.
- Describe how HTTP requests and responses work, and explain the role of an API in exchanging data.

When you click a link or press “search,” your device does not hold the whole website — it asks a distant computer for it. Your browser is the client that makes the request; a server somewhere answers with a response. This exchange, governed by shared rules called protocols, is how every web application communicates. Knowing how requests, responses, and APIs work explains what happens between your click and the page you see.

? Guiding question: How do the client and server communicate to make a web application work, and what rules and messages do they use?

Key Fact:
Web applications work by a client sending requests and a server returning responses, using protocols (HTTP/HTTPS) and often APIs that exchange data in formats like JSON.

Key Concepts:

- client / server
- HTTP / HTTPS
- HTTP method
- status code
- API / JSON

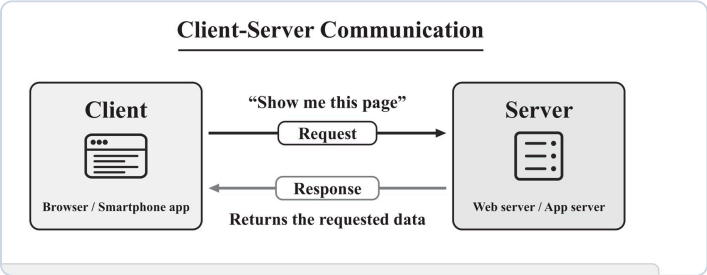
Point!

1. Client and Server

(1) **(Client)**…the side that requests data or processing from the server. Web browsers and smartphone apps fall into this category.

(2) **(Server)**…the side that receives requests from clients, processes them, and returns responses.

(3) A web application operates by the client sending a request and the server returning a response.



Client-Server Communication

KEY TERMS
client — the side that requests data or processing.
server — the side that processes requests and returns responses.



Many clients request; one server responds.

2. HTTP and HTTPS

(1) **(HTTP (HyperText Transfer Protocol))**…a communication protocol for exchanging data between clients (browsers and apps) and servers.

(2) **(HTTPS (HTTP Secure))**…a communication protocol that adds TLS encryption to HTTP. Because communication content is encrypted, the risk of its being read by third parties is reduced.

(3) Websites whose URLs begin with “https://” use HTTPS communication.

3. HTTP Requests and Responses

(1) Requests sent from the client to the server include information called an HTTP method that indicates “what kind of operation is desired.”

PAUSE & THINK

Why should a login page use HTTPS rather than plain HTTP?

Method	Role	Examples
GET	Retrieve data from the server	Displaying a web page, retrieving search results
POST	Send data to the server	Sending login information, submitting form contents

(2) Responses returned from the server to the client include a number indicating the result of processing, called a status code. Some typical examples are as follows.

Status code	Meaning
200	Success (the request was processed successfully)
404	Page not found (the specified URL does not exist)
500	Server error (a problem occurred on the server side)

4. API (Application Programming Interface)

(1) **(API)**…a system for exchanging data between different pieces of software.

(2) In web applications, the frontend often sends requests to the backend through an API and receives data.

(3) Data exchanged through an API is often described in a format called **(JSON (JavaScript Object Notation))**. JSON is a data format that is both easy for humans to read and for computers to process.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A The client is the side that requests data or processing from the server, and the server is the side that returns the response.

B HTTPS is a communication protocol that adds encryption to HTTP.

C The GET method is for sending data to the server.

D Status code 404 indicates that the page is not found.

(2) Match each description (a–c) with the most appropriate term from the choices below (A–C).

a A communication protocol for exchanging data between web browsers and servers

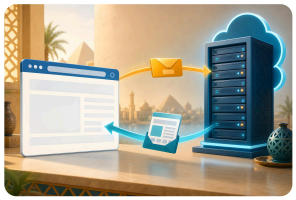
b A system for exchanging data between different pieces of software

c A communication protocol that adds encryption to HTTP

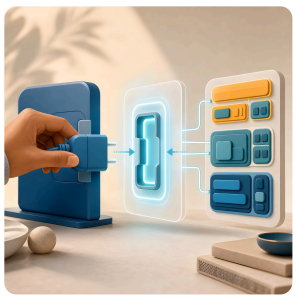
[Choices] **A** HTTP **B** API **C** HTTPS

KEY TERMS

HTTP method — says what is wanted (GET, POST).
status code — reports the result (200, 404, 500).



One HTTP request, one matching response.



An API is a fixed contract for exchanging data.

✓ Solution

(1)

A Correct description of the roles of client and server. Therefore, ○.

B HTTPS adds SSL/TLS encryption to HTTP. Therefore, ○.

C The GET method is for retrieving data from the server and the POST method is for sending data. Therefore, ×.

D 404 is a status code indicating that the page is not found. Therefore, ○.

(2)

a: The web communication protocol is HTTP. **A**

b: Data exchange between pieces of software is API. **B**

c: HTTP with added encryption is HTTPS. **C**

Try

1 Answer the following questions.

1. What is the term for the side that requests data or processing from the server (e.g., a web browser)?
2. What is the term for the side that receives requests from clients, processes them, and returns responses?
3. What is the term for the communication protocol that adds encryption to HTTP?
4. What is the term for the HTTP method for retrieving data from the server?
5. What is the term for the system for exchanging data between different pieces of software?

2 Answer the following questions.

(1) From the following options (A–D), choose the combination of status code and meaning that is NOT appropriate.

A 200 — The request was processed successfully

B 404 — Page not found

C 500 — The request was processed successfully

D 500 — A problem occurred on the server side

EXAM-STYLE QUESTION

Explain what happens, step by step, when a browser requests a web page and the server responds. **[6] Refer to:** client, request, HTTP method, response, status code.

PAUSE & THINK

If a browser receives status code 404, what should it tell the user — and whose side had the problem?

(2) From the following options (A–D), choose the one that most appropriately describes an API.

A A technology for adjusting the appearance of web pages

B A system for exchanging data between different pieces of software

C A technology for encrypting data and sending it safely

D An HTTP method for retrieving data from the server

⚙ Think as an Engineer — Investigate & Decide

A school in your city wants a web app where students log in, view their timetable, and download a PDF. You plan how the client and server communicate.

☐ **Send everything over plain HTTP**

☐ **Use HTTPS and the right method for each action**

1 · Choose the method. For three actions — loading a student's grade history, submitting the login form, and requesting today's schedule from another school system — say whether GET or POST fits, and for the last say whether an API is involved.

2 · Choose the protocol. Decide whether the login should use HTTP or HTTPS, and give the reason.

3 · Decide. Recommend the communication design and give two reasons. **In pairs,** compare your method choices and agree together on which action must use POST.

Give evidence for your answer.

Stuck? GET retrieves and shows; POST sends data such as a login; HTTPS encrypts so passwords are not intercepted.

🌍 In a New Context

A weather service offers an API that returns today's forecast when a client sends a request. Using what you learned, name which HTTP method the client would use to retrieve the forecast, why the service should use HTTPS, and one advantage of receiving the data as JSON.

KEY TERMS

API — lets different software exchange data.

JSON — a readable data format.

❓ Lesson Question — Answered

The lesson asked how the client and server communicate and what rules and messages they use. A web application works when a client — a browser or app — sends a request and a server processes it and returns a response. They communicate through a protocol: HTTP carries the messages, and HTTPS adds SSL/TLS encryption so the content is not easily intercepted. Each request carries an HTTP method that says what is wanted — GET to retrieve, POST to send — and each response carries a status code that reports the result, such as 200 for success or 404 for not found. Often the frontend and backend exchange data through an API, frequently in the JSON format, which is easy for both people and computers to read. Overall, communication in a web application is an organized exchange of requests and responses, shaped by protocols, methods, status codes, and APIs.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

A web application operates by communication between a (a), which requests data or processing from the server, and a (b), which receives the request, processes it, and returns a response. This communication uses a protocol called (c). The protocol that adds SSL/TLS encryption to (c) is called (d).

1. Fill in the blanks (a)–(d).

3 Answer the following questions.

(1) For each of operations (a–d), indicate which HTTP method is used: **A** GET, **B** POST.

a Displaying a web page

b Entering an ID and password into a login form and submitting them

c Displaying search results

d Submitting the contents of a contact form

(2) From the following options (A–D), choose the one that most appropriately describes HTTPS.

A A method for retrieving data from the server

B A system for exchanging data between different pieces of software

C A communication protocol that adds SSL/TLS encryption to HTTP

D A status code indicating that a problem has occurred on the server side

⭐ REFLECT & CHALLENGE

Reflect: When a page fails to load, which status code would help you most to understand why, and what would it tell you?
Challenge: Name one app you use and describe one request it sends and the response it expects.

⭐ Key takeaway

Remember:
Client requests, server responds — over HTTP/HTTPS, with a method (GET/POST), a status code, and often an API returning JSON.



Learning Objectives

- Explain the distinct roles of HTML, CSS, and JavaScript in building a web page.
- Describe how semantic HTML, responsive design, and frameworks improve a modern frontend.

Open any web page and you are seeing three technologies at work together. HTML builds the structure — the headings, paragraphs, and links, like the frame of a building. CSS adds the appearance — colors, layout, and fonts, like a building's interior. JavaScript adds behavior — search boxes that respond, buttons that count, screens that update. A modern frontend goes further: it makes pages meaningful, adaptable to every screen, and faster to build.

? Guiding question: How do HTML, CSS, and JavaScript work together to build a web page, and what makes a modern frontend meaningful, adaptable, and efficient to build?

Key Fact:
A web page is built from HTML (structure), CSS (appearance), and JavaScript (behavior); a modern frontend adds semantic HTML, responsive design, and frameworks.

Key Concepts:

- HTML / CSS / JavaScript
- semantic HTML
- accessibility / SEO
- responsive design
- framework

KEY TERMS
HTML — structure.
CSS — appearance.
JavaScript — behavior.



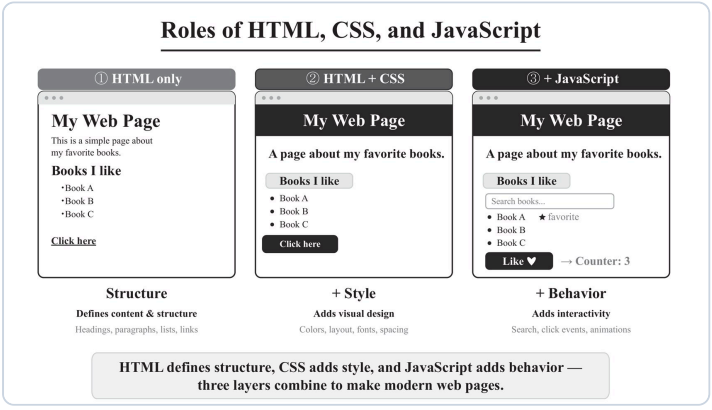
HTML structures, CSS styles, JS adds behavior.

Point!

1. The Roles of HTML, CSS, and JavaScript

(1) Web pages are mainly based on the following three technologies.

Technology	Role	Analogy
HTML	Defines the structure (skeleton) of a web page. Describes elements such as headings, paragraphs, images, and links.	The frame of a building
CSS	Specifies the appearance of a web page (colors, layout, size, fonts, etc.).	The interior and exterior of a building
JavaScript	Adds behavior and interactivity to a web page. Activates responses when buttons are pressed and updates only part of the screen.	The electrical and plumbing systems of a building



Roles of HTML, CSS, and JavaScript

(2) These three technologies each have independent roles, and a web page is created using all of them in combination.

2. Semantic HTML

(1) **(Semantic HTML)**…the concept of describing each part of a web page using element names (tags) that indicate their meaning.
e.g., Use <header> at the top of the page, <nav> for navigation, <main> for the body, and <footer> at the bottom of the page.

(2) Benefits of using semantic HTML

① Improved **(accessibility)**…screen reader software can correctly understand the structure of the page. Information becomes easier to convey to users with visual impairments.

② Improved **(SEO (Search Engine Optimization))**…search engines can more easily understand the content of a web page correctly, making it more likely to be displayed appropriately in search results.

3. Responsive Design

(1) **(Responsive design)**…a design concept in which the layout is automatically adjusted according to different screen sizes such as PCs, smartphones, and tablets.

(2) With a single web page, a display that is easy to view from any device can be achieved.

(3) As of 2026, websites are frequently accessed from smartphones, and support for responsive design is widely required. Against this backdrop, the **(mobile-first)** concept—prioritizing small screens such as smartphones from the design stage—has also become widespread.

4. Frameworks

(1) **(Framework)**…a structure that provides commonly used functions and components in advance, in order to make web application development more efficient.

(2) By using a framework, developers do not need to write all code from scratch and can develop more efficiently.

(3) Major frontend frameworks

Framework name	Characteristics
React	Developed by Meta (formerly Facebook). It can efficiently update only part of the screen and is suited to large-scale web applications.
Vue	Has a simple structure and allows web applications to be built incrementally from small features.
Next.js	In addition to React's features, includes mechanisms to speed up page display.

PAUSE & THINK

On a page you use often, which parts are structure (HTML), which are style (CSS), and which react to you (JavaScript)?



Responsive design re-flows to each screen size.

KEY TERMS

responsive design — layout adapts to screen size.
framework — ready-made functions for efficient development.



Frameworks assemble pages from ready-made parts.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A HTML is a technology that defines the structure (skeleton) of a web page.

B CSS is a technology that adds behavior to a web page.

C Responsive design is a concept in which the layout is automatically adjusted according to different screen sizes.

D Using a framework eliminates the need to write all code from scratch and allows efficient development.

(2) Match each role (a–c) with the most appropriate technology from the choices below (A–C).

a Defines the structure of a web page

b Specifies the appearance of a web page

c Adds behavior and interactivity to a web page

[Choices] **A** HTML **B** CSS **C** JavaScript

✓ Solution

(1)

A HTML defines the structure of a web page. Therefore, ○.

B CSS is a technology that specifies the appearance of a web page. Adding behavior is JavaScript. Therefore, ×.

C Correct definition of responsive design. Therefore, ○.

D Correct description of the benefit of frameworks. Therefore, ○.

(2)

a: Defining structure is HTML. **A**

b: Specifying appearance is CSS. **B**

c: Adding behavior is JavaScript. **C**

PAUSE & THINK

Why does a screen reader understand a page better when it uses <header>, <nav>, and <main> instead of plain boxes?

EXAM-STYLE QUESTION

Explain how HTML, CSS, and JavaScript each contribute to a library page that lists books, is styled, and filters the results as the user types. **[6] Refer to:** structure, appearance, behavior.

Try

1 Answer the following questions.

1. What is the four-letter abbreviation for the technology that defines the structure (skeleton) of a web page?
2. What is the three-letter abbreviation for the technology that specifies the appearance of a web page (colors, layout, size, etc.)?
3. What is the term for the technology that adds behavior and interactivity to a web page?
4. What is the term for the design concept in which the layout is automatically adjusted according to different screen sizes?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that is NOT an appropriate benefit of using semantic HTML.

A Screen reader software can correctly understand the structure of the page.

B Search engines can more easily understand the content of a web page.

C The display speed of a web page necessarily becomes faster.

D Information becomes easier to convey to users with visual impairments.

(2) From the following options (A–D), choose the one that most appropriately describes a framework.

A A language for defining the structure of a web page

B A structure for making web application development more efficient

C A technology for encrypting data and sending it safely

D A design concept for supporting different screen sizes

PAUSE & THINK

Which of the benefits below is about how the page looks, and which is about how well others can find and read it?

Think as an Engineer — Investigate & Decide

A student club wants a simple web page that lists its events, is styled in the club's colors, and has a button showing how many people have signed up.

☐ **Write everything in one file**

☐ **Use HTML, CSS, and JavaScript for their proper roles**

1 · Observe first. Open one web page you use often and note two things on it that change when you click or type. Then, for three features of the club page — the list of events, the club's colors and layout, and the live sign-up counter — say which technology (HTML, CSS, or JavaScript) does each, and why.

2 · Make it reach everyone. Name one change that would make the page easier to read on a phone (responsive design), and one semantic-HTML choice that would help a screen reader.

3 · Decide. Recommend whether the club should separate HTML, CSS, and JavaScript, and give two reasons. **In pairs,** compare your role assignments and agree together on what the sign-up counter needs.

Give evidence for your answer.

Stuck? HTML holds the content; CSS styles it; JavaScript makes it react and update.

🌍 In a New Context

A news site must look good on phones and be found easily by search engines. Using what you learned, name one responsive-design choice and one semantic-HTML choice it should make, and say what each one improves.

❓ Lesson Question — Answered

The lesson asked how HTML, CSS, and JavaScript work together and what makes a modern frontend strong. A web page is built from three technologies with independent roles: HTML defines the structure, CSS specifies the appearance, and JavaScript adds behavior and interactivity — combined, they create the page. A modern frontend goes further: semantic HTML uses meaningful tags so screen readers and search engines understand the page (improving accessibility and SEO); responsive design adjusts the layout to any screen size, with a mobile-first approach; and frameworks such as React, Vue, and Next.js provide ready-made functions so developers build efficiently without writing everything from scratch. A modern web page is therefore not one technology but a combination — structured, styled, interactive, meaningful, adaptable, and efficient to build.

Exercise

KEY TERMS

accessibility — usable by people with impairments.
SEO — being found by search engines.

⭐ REFLECT & CHALLENGE

Reflect: Of structure, style, and behavior, which do you notice most when a web page is badly made, and why? **Challenge:** Name one website you use and describe one thing HTML, CSS, and JavaScript each do on it.

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

Web pages are mainly composed of three technologies: (a), which defines the structure of the page; (b), which specifies the appearance; and (c), which adds behavior and interactivity. The concept of adjusting the layout according to different screen sizes such as PCs, smartphones, and tablets is called (d).

1. Fill in the blanks (a)–(d).

3 Answer the following questions.

(1) For each of tasks (a–f), indicate which technology is used to achieve it: **A** HTML, **B** CSS, **C** JavaScript.

a Describing the structure of headings and paragraphs

b Changing the color and size of text

c Updating part of the screen when a button is pressed

d Placing images and links

e Setting the background color of the entire page

f Validating the contents entered into a form

(2) From the following options (A–D), choose the one that most appropriately describes the reason responsive design is needed.

A To improve the display speed of websites

B Because websites are accessed from various devices such as PCs, smartphones, and tablets

C To raise rankings in search engines

D To add behavior to web pages

Key takeaway

Remember:
HTML structures, CSS styles, JavaScript behaves — and semantic HTML, responsive design, and frameworks make a modern frontend meaningful, adaptable, and efficient.